

MGT 456: Innovative Product and Program Management (iPM) (aka/formerly: Digital Product Management at UCSD & MIT)

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Office Hours: By Appointment
Teaching Assistant: TBD; Office Hours: TBD

COURSE OBJECTIVES

The Innovative Product and Program Management (iPM) class deals with the knowledge, skills, and competencies needed to become a creative, high-potential Product and Program Manager in the AI age. While we focus on increasingly ubiquitous digitally-enriched products and projects, we also consider management of conventional products and programs (life sciences and electromechanical offerings) in a digital, high-velocity environment. Key aspects of product strategy, portfolio and program management, agile product development, and product lifecycle management are discussed, as well as the associated responsibilities of the product manager. This course is intended for students seeking a PM position in either a new or established enterprise and is based on management development programs developed for top companies by the Professor and taught at MIT and UC San Diego. The course augments and enriches content on business strategy, marketing, entrepreneurship, and operations management- so counts towards Marketing and Supply Chain certificates at UC San Diego's Rady School.. Class sessions consist of instructor-led discussions, in-class exercises and guest speakers on PM topics. Students work on team-based exercises, and a team-based hands-on PM project in the field with a silicon valley company.

PERFORMANCE EVALUATION

The final grade in this class will be based on your performance as follows:

Homework/Case Critique Assignments	15%	(Team)
Class Participation	25 %	(every session, Individual)
Midterm Exam (Week 7/8; In-class)	30%	(Individual)
Class Project with intermediate deliverables	30%	(Team)
Total	100%	

Homework/Case Critique Assignments (15%):

A combination of team papers and individual homeworks (briefs, as announced before hand) and submitted through the Canvas dropbox, these assignments will help practice critical and creative thinking skills needed for product management.

Class Participation (25%):

Your class participation (contribution to class discussions and/or response to in-class questions/quizzes) will count for 25% of the class grade. We may also use Zoom/canvas online



discussion forums for class participation. You are expected to be an active participant throughout the entire class and to contribute to the quality of the discussion. Please note that the frequency (i.e., the quantity) of your interventions in class is not a key criterion for effective class contribution.

Some criteria used to evaluate class contribution are as follows:

1. Is the participant deeply engaged in class? **Use of laptops and electronic devices for non-class purposes can negatively affect class participation and learning experience.**
2. Is the participant a good listener? Are the points made relevant to the discussion? Are they linked to the comments of others? Do the comments show evidence of analysis of the case?
3. Is there willingness to test new ideas, or are all comments "safe"? (For example, repetition of case facts without analysis and conclusions).
4. Do comments clarify or build upon the important aspects of earlier comments and lead to a clearer understanding of the case?

Note: You are expected to attend EVERY class session; you are responsible for the material covered in class whether or not you attend.

Midterm Exam (Week 7/8; In-class) 30% (Individual)

There will be an in-class exam on Week 7/8, whose goal is to review the material learned and learn to apply them in real-life settings. The exam will be partly analytical and the exact modality will be discussed in class two weeks prior to the exam.

Class Project (with Intermediate Milestones): 30%

The class project is a key element of this class, in which you will focus as a team on a digital product you like, understand the roadmap and product strategy, identify unmet needs and define a key feature not yet offered in the product, develop requirements, complete a testable prototype, and collect customer feedback. You will learn to use AI-based tools introduced in the class. At the mid-term milestone, each team will submit their product strategy/roadmap and unmet needs. There will be a project deliverable due the last class session which will count for 20% of the class grade, and will include a prototype tested with user testing resources. As a team, you would present your project test results and lessons learned. There is no final exam in this class.

COURSE MATERIALS

The class is mainly an action-oriented class with reference articles, books, and videos. A small collection of cases are available through a digital course reader that can be downloaded from the Harvard web site using links provided in the Canvas Syllabus page.

A number of digital articles and videos are assigned for every class through the syllabus below. In addition, the course intranet page ("Canvas") may contain additional articles and videos on the topic and descriptions of technologies from various sources. Product Management is a rapidly



growing field with a number of great reference books, podcasts and webinars. The following books are recommended as reference, in addition to the material mentioned in the Canvas site:

Inspired: How to Create Tech Products that Customers Love, by Marty Cagan, Silicon Valley Product Group, [Book](#), [Video](#)

[Shipping Greatness: Practical lessons on building and launching outstanding software, learned on the job at Google and Amazon](#), by Chris Vander Mey- Webcast available [here](#).

[Cracking the PM Interview: How to Land a Product Manager Job in Technology](#), by Gayle Laakmann McDowell and Jackie Bavaro, offers a good discussion of careers- [video here](#).

A growing number of useful videos are available, including [Lean Product Management](#), [Product School channel](#), [This Is Product Management](#), [Product Collective](#), etc.

ACADEMIC INTEGRITY: As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The UCSD Policy on Integrity of Scholarship can be viewed [here](#).

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member. The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.** Please visit the [OSD website](#) for further information or contact Office for Students with Disabilities.

COURSE SCHEDULE

The detailed course outline provided below lists the readings for each class. **I expect that you will read the assigned materials in advance and prepare a careful analysis before class.** You and I will work together to create the best learning environment that we can. I will schedule extra review sessions as the need arises. In the spirit of continuous improvement, I will seek informal feedback several times during the quarter. Please let me know throughout the quarter, individually or collectively, if there is anything I can do to make this class better for you!



COURSE SCHEDULE/TOPIC SEQUENCE

<i>Session 1</i>	INTRODUCTION TO PRODUCT AND PROGRAM MANAGEMENT: CAREER SPECTRUM, ROLES, RESPONSIBILITIES, DISTINCTIONS Creative and Analytical Dimensions of Product Management Executive overview/Sprint of the Topics
<i>Session 2</i>	CREATING AND EVOLVING PRODUCT STRATEGY AND VISION INTRODUCTION TO ROADMAPPING & PORTFOLIO MANAGEMENT
<i>Session 3</i>	PRODUCT STRATEGY, ARCHITECTURE & ROADMAPPING INTRO TO PRODUCT DEFINITION AND OPPORTUNITY IDENTIFICATION
<i>Session 4</i>	PRODUCT POSITIONING & PRIORITIZATION; PRODUCT STACK BUILDING; INTRODUCTION TO USER/CUSTOMER STUDIES
<i>Sessions 5</i>	PRODUCT DESIGN, PROTOTYPING (WIREFRAMING) AND VALIDATION; Introduction to Product and Project Risk Management
<i>Session 6</i>	Agile Product and Program Management
<i>Session 7</i>	In-class Exam and Review of User testing Plans
<i>Session 8</i>	Product Lifecycle Mgmt. and Pricing
<i>Session 9</i>	Meetings with TA and Professor- By appt.- no class on Memorial Day
<i>Session 10</i>	PRESENTATIONS AND SYNTHESIS/WRAP-UP (No Final Exam)



DETAILED SYLLABUS (BEING UPDATED – TO BE POSTED ON CANVAS)

Session 1	INTRODUCTION TO DIGITAL PRODUCT MANAGEMENT: ROLES, RESPONSIBILITIES, ACCOMPLISHMENTS PURPOSEFUL PRODUCTS, IMPACTFUL JOURNEY What is a Product Manager?, https://www.quora.com/What-is-a-product-manager-What-do-you-study-in-order-to-become-one-What-skills-do-you-need What is a Product?, https://www.youtube.com/watch?v=z18Gpcu5Tyg What's the difference between a Project Manager and a Product Manager? How the Software Industry Redefines Product Management Caveats with Product Thinking: Ensuring Customer/Solution Orientation Product Manager (PM) has become a pivotal role in many growth and value creation-driven organizations. We will discuss the reason for this trend as well as the different flavors, responsibilities, and expectations.
Session 2 In-class case studies	CREATING AND EVOLVING PRODUCT STRATEGY AND VISION INTRODUCTION TO ROADMAPMING & PORTFOLIO MANAGEMENT Project Deliverable: Submit Your Project Team and Candidate Product Choices Considered What It Takes to Become a Great Product Manager, Julia Austin, DECEMBER 13, 2017, https://hbr.org/2017/12/what-it-takes-to-become-a-great-product-manager Amazon's approach to Product Management, https://www.quora.com/Amazon-company-What-is-Amazons-approach-to-product-development-and-product-management Building Products with a Purpose What is Strategy? http://www.fastcompany.com/42485/michael-porters-big-ideas. Products to Platforms W. Chan Kim and Renée A. Mauborgne, "Creating New Market Space." Innovation Is as Much About Finding Partners as Building Products Disruptive Innovation , Interview Additional references: Wireframing Your Web Application



	What is the difference between program, product, and project manager roles? How can an entry level Product Manager become a Kickass Product Manager? How to Build a Data Analytics Strategy by LinkedIn Product Manager. https://www.youtube.com/watch?v=Xc5HrS3MTz0
Session 3	PRODUCT ARCHITECTURE & ROADMAPING; CUSTOMER INTERVIEWS INTRO TO PRODUCT DEFINITION AND OPPORTUNITY IDENTIFICATION Case due: INTUIT QUICKBOOKS (see digital course packet; refer Canvas for questions) Readings: 12 Tips for Early Customer Development Interviews Doing Market Research with Google's Keyword Tool 5 Ways to Design Products Customers Love “Finding Your Innovation Sweet Spot,” Harvard Business Online. How Gatorade Invented New Products by Revisiting Old Ones What AI can and can’t do (yet) for your business, McKinsey Quarterly Videos: Customer Discovery: Steps Good and Bad Customer Interviews Building Ethical and Persuasive Products Design Thinking: A Quick Introduction Opportunity Recognition and Leveraging Why Analytics Is Key for Product Managers by Airbnb Data Scientist
Session 4	PRODUCT DESIGN & INTRODUCTION TO USER EXPERIENCE Introduction to Usability Developing Customer Personas; Product Manager Personas Project Deliverable: Submit Your Product Choice and Customer Data/Market Opportunity Profile using Template Readings: Writing a Good Product Requirements Document (PRD) Elements of User Experience 10 Tech Terms to Know Model-View-Controller (MVC) in web programming Guide to IFttt Native App or Mobile Web How to ask customers questions? The 5 best guides for creating customer personas UserTesting Blog Video: Product Requirements Document (PRD) Driving Discovery and Disruption What Customers Want from Your Products “Finding Your Innovation Sweet Spot,” Harvard Business Online



<i>Sessions 5</i>	PRODUCT DESIGN, PROTOTYPING (WIREFRAMING) AND VALIDATION Case due: DROPBOX: 'IT JUST WORKS' (see digital course packet; refer Canvas Dropbox for questions to answer in your case analysis) <u>A REFRESHER ON A/B TESTING, HTTPS://HBR.ORG/2017/06/A-REFRESHER-ON-AB-TESTING</u> <u>Yes, A/B Testing Is Still Necessary, https://hbr.org/2014/12/yes-ab-testing-is-still-necessary</u> <u>A/B Testing Lessons Learned</u> <u>A/B Testing and the Benefits of an Experimentation Culture</u>, Wyatt Jenkins <u>Wireframing Your Web Application</u>
<i>Session 6</i>	PRODUCT POSITIONING & PRIORITIZATION; PRODUCT STACK BUILDING <u>Making Design Trade-offs</u> <u>Product Prioritization Techniques</u> <u>Slack's Method for Processing User Feedback</u> <u>Wireframing Your Web Application</u> Project Deliverable: Submit Your Product Requirements Document and Preliminary Wireframe to be Tested for Discussion with the Professor
<i>Session 7</i>	Product Management Interface with Projects; Agile Product Management Case: Product Development at Dell (see digital course packet; refer Canvas Dropbox for questions to answer in your case analysis) Project Deliverable: Submit Your Plan and Progress for User-Testing of Your Wireframe Readings: <u>Speed Up Your Product Development Without Losing Control</u> <u>Six Myths of Product Development</u> <u>Agile Product Ownership in a Nutshell</u>



Session 8	Product Lifecycle Mgmt; Pricing Case: Basecamp Product Launch Strategy Planning the Product Launch Introduction to Lifecycle Pricing Why Analytics Is Key for Product Managers by Airbnb Data Scientist, https://www.youtube.com/watch?v=sNiQXszVr9Y Using Analytics to Build Better Products - Product Analytics Summit - Spenser Skates, https://www.youtube.com/watch?v=KtufIU FouTU What We Learned from Tracking 1 Trillion User Actions - Product Analytics Summit
Session 9/10 PITCH	Launching and Pitching Products PROJECT PRESENTATIONS AND WRAP-UP Pitching and Presenting Projects Future of Product Management Planning Your Product Management Career NO CLASS Session 9 p(DUE TO MEMORIAL DAY)

